ICT 4442 - Deep Learning

Mini Project Synopsis

Comparative Analysis of Deep Learning Architectures for Sentiment Analysis of Movie Reviews

Submitted By

Meesha Gupta - 230953090
Ratul Jha - 230953338
Harshavardhan Reddy - 230953396
Kshitij Singh - 230953460

SCHOOL OF COMPUTER ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY,
MANIPAL ACADEMY OF HIGHER EDUCATION

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Brief Description of the Project

Sentiment analysis is the task of automatically identifying the opinion expressed in text. It is a well studied problem in natural language processing with applications in product review monitoring and social media analytics. For movie reviews, the goal is to determine whether the reviewer expresses a positive or negative sentiment.

This project focuses on binary sentiment classification of movie reviews using the IMDb Large Movie Review Dataset. The core objective is to understand how different deep learning architectures approach the same problem. Four architectures are implemented and compared: a Multi-Layer Perceptron (MLP), a one dimensional Convolutional Neural Network (1D-CNN), a Bidirectional Long Short Term Memory network (BiLSTM), and a fine tuned DistilBERT Transformer model.

These four models represent distinct families of deep learning architectures. The MLP serves as a simple feed forward baseline that processes text as a bag of averaged word embeddings. The 1D-CNN applies convolutional filters over word sequences to detect local phrase level patterns. The BiLSTM introduces sequential processing with gated memory to retain contextual information across long distances. DistilBERT is a compact variant of the BERT Transformer, which uses self attention to model relationships between words simultaneously.

All four models are evaluated on the same dataset split, using the same preprocessing pipeline and the same evaluation metrics. This ensures any observed performance differences can be attributed to architectural choices. The project also includes overfitting analysis, learning curve inspection, and error analysis on misclassified examples.

Datasets Identified

Dataset: IMDb Large Movie Review Dataset (Maas et al., 2011)
Source: `https://ai.stanford.edu/~amaas/data/sentiment/`

The dataset contains 50,000 movie reviews collected from the IMDb platform, each labelled as either positive or negative. The class distribution is balanced at 25,000 reviews

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per class. The official release provides 25,000 reviews for training and 25,000 for testing, with an additional 50,000 unlabelled reviews that are not used in this project.

The training split of 25,000 reviews is divided into 20,000 samples for training and 5,000 for validation. The official test split of 25,000 reviews is kept completely separate and used only for the final evaluation. No hyperparameter decisions, model selection, or threshold tuning are based on test set performance.

Proposed Models for Comparison

1. **Multi-Layer Perceptron (MLP):** A standard feed-forward neural network acting as a foundational baseline. In this architecture, the text is represented as a bag of averaged word embeddings, completely discarding word order. The continuous representation is passed through dense hidden layers with non-linear activations (e.g., ReLU) before a final sigmoid classification layer. This model establishes the baseline performance possible without explicit sequence modelling.
2. **1D Convolutional Neural Network (1D-CNN):** A convolutional architecture designed to capture local word patterns and short phrase-level features. It applies 1D convolutional filters of varying sizes (acting as n-gram detectors) across the sequence of word embeddings. This is followed by a global max-pooling layer to extract the most salient features regardless of their position in the text, making it highly effective at detecting strong sentiment-bearing phrases.
3. **Bidirectional Long Short-Term Memory (BiLSTM):** A recurrent neural network architecture that explicitly models the sequential nature of text. By processing the sequence of embeddings in both forward and backward directions, the BiLSTM captures both past and future context for every word. The gated memory cells prevent the vanishing gradient problem, allowing the network to retain long-range dependencies that are critical for understanding complex sentiment in lengthy movie reviews.
4. **DistilBERT:** A highly capable, fine-tuned pretrained Transformer model. As a distilled version of BERT, it utilizes multi-head self-attention mechanisms to model complex, bidirectional relationships between all words simultaneously. Because it is pretrained on massive text corpora, it brings a deep prior understanding of language syntax and semantics, enabling it to handle difficult edge cases like sarcasm, subtle negation, and mixed sentiments far better than models trained entirely from scratch.

Model-to-Member Assignment

1. Meesha Gupta (230953090): MLP
2. Ratul Jha (230953338): 1D-CNN
3. Harshavardhan Reddy (230953396): DistilBERT
4. Kshitij Singh (230953460): BiLSTM

Existing State-of-the-Art Methods Related to the Proposed Project

The current state of the art in sentiment analysis relies on large pretrained Transformer based language models. BERT demonstrated that deep bidirectional attention over large

pretrained representations outperforms task specific recurrent and convolutional models across many NLP benchmarks. On the IMDb dataset, BERT family models routinely achieve accuracies above 94 percent, with some fine tuned variants reaching beyond 96 percent. DistilBERT is a knowledge distilled version of BERT that retains approximately 97 percent of BERT's performance while being 40 percent smaller and 60 percent faster. Prior to Transformers, CNN based approaches and BiLSTM models represented the competitive baselines in sentiment analysis. While these architectures perform reasonably well on IMDb, they fall short of Transformer based methods.

Objectives

1. To implement and train four deep learning architectures (MLP, 1D-CNN, BiLSTM, and DistilBERT) for binary sentiment classification on the IMDb movie review dataset under a common preprocessing and evaluation protocol.

2. To systematically compare the performance of these architectures using accuracy, precision, recall, F1-score, and AUC, and to analyse how architectural differences influence classification outcomes.

3. To conduct error analysis on misclassified examples in order to identify patterns such as negation, sarcasm, and mixed sentiment that challenge each model differently.

Importance / Motivation of the Proposed Project

Sentiment analysis of movie reviews is a benchmark task that allows controlled comparison of NLP models. The IMDb dataset is widely used in the research community, so the results are directly comparable to published work. The project compares these architectures systematically by holding everything else constant, including the dataset, the split, the preprocessing, the metrics, and the regularization strategy. This produces an interpretable comparison of four different approaches to text classification. From a learning perspective, the project covers the core deep learning paradigms taught in the course. Each team member implements and takes ownership of one architecture to ensure individual understanding of each approach.

Work Plan

Sl. No.	Component / Work Element / Milestone	Expected Completion
1	Dataset Acquisition and Preprocessing: Download IMDb dataset, inspect, clean, split, build vocabulary	1 Sep 2026
2	MLP Implementation: Build and train MLP baseline, log training and validation curves	8 Sep 2026
3	1D-CNN Implementation: Build and train 1D-CNN, compare with MLP on validation set	12 Sep 2026
4	BiLSTM Implementation: Build and train BiLSTM, integrate with shared evaluation pipeline	18 Sep 2026
5	Interim Report Submission: Submit Part B with literature review, preprocessing code, and model results	25 Sep 2026
6	DistilBERT Fine-tuning: Fine-tune DistilBERT, tune learning rate, monitor overfitting	5 Oct 2026
7	Final Evaluation: Run all four models on test set, generate metrics, confusion matrices, ROC curves	14 Oct 2026
8	Error Analysis and Discussion: Analyse misclassified examples, document patterns and explanations	18 Oct 2026
9	Report Writing: Complete final report (Part C), integrate all sections	27 Oct 2026
10	Presentation and Code Submission: Prepare slides, rehearse viva, push final code to GitHub	31 Oct 2026

Preliminary Reference List

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